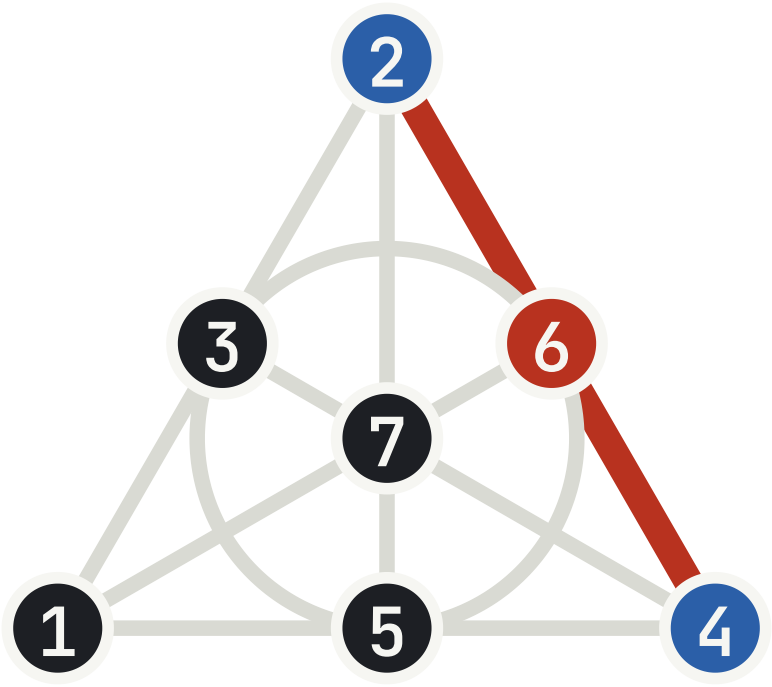
FANO PLANE · GF(2)³ · 7 POINTS · 7 LINES

Every file is one constant

Each SVG you made is read as text, its marked dots are located, and the result becomes a 7-bit mask: bit `n-1` is on when point `n` is marked. Your numbering is the XOR numbering, so on every line `a ⊕ b ⊕ c = 0`.

Pick two points

The third point on their line is computed as `a ⊕ b`, then the line is lit. Use Tab and Enter, or tap.



READOUT

2 ⊕ 4 = 6

A	010	2	0x2
B	100	4	0x4
A ⊕ B	110	6	0x6
A⊕B⊕C	000	0	0x0

Clear selection

THE SEVEN LINES, CHECKED

LINE	A	B	C	A⊕B⊕C
1-3-2	001	011	010	000
1-5-4	001	101	100	000
1-7-6	001	111	110	000
2-6-4	010	110	100	000
2-7-5	010	111	101	000
3-5-6	011	101	110	000
4-7-3	100	111	011	000

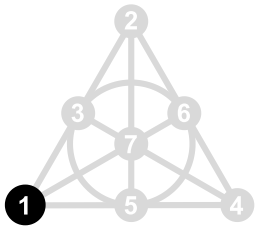
Your files

Click once to set View A, again on another file for View B. The mask under each image comes from the file's text, not from its name.

Fano_empty.svg

0000000 · 0x00

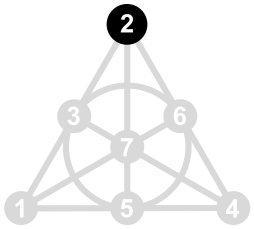
NO MARKED POINTS



Fano_point_1.svg

0000001 · 0x01

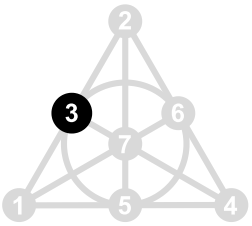
POINTS 1



Fano_point_2.svg

0000010 · 0x02

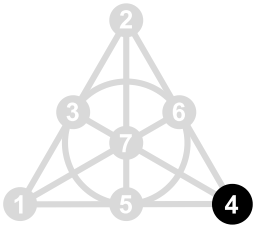
POINTS 2



Fano_point_3.svg

0000100 · 0x04

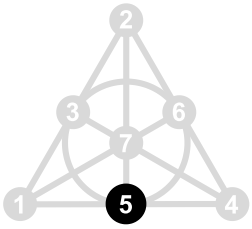
POINTS 3



Fano_point_4.svg

0001000 · 0x08

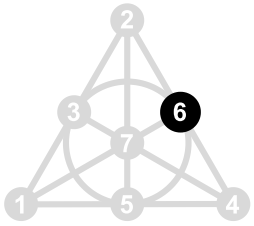
POINTS 4



Fano_point_5.svg

0010000 · 0x10

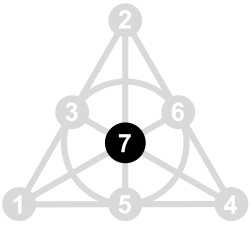
POINTS 5



Fano_point_6.svg

0100000 · 0x20

POINTS 6



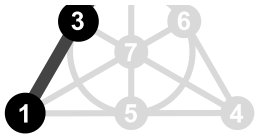
Fano_point_7.svg

1000000 · 0x40

POINTS 7



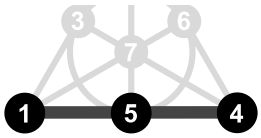




Fano_line_1_3_2.svg

0000111 · 0x07

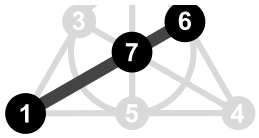
POINTS 1,2,3



Fano_line_1_5_4.svg

0011001 · 0x19

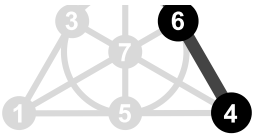
POINTS 1,4,5



Fano_line_1_7_6.svg

1100001 · 0x61

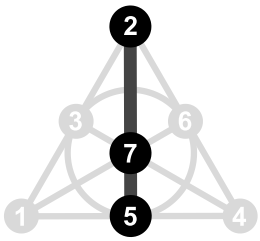
POINTS 1,6,7



Fano_line_2_6_4.svg

0101010 · 0x2a

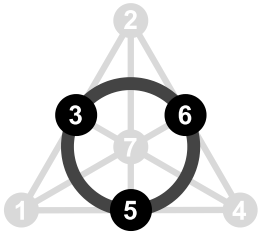
POINTS 2,4,6



Fano_line_2_7_5.svg

1010010 · 0x52

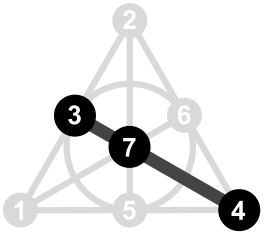
POINTS 2,5,7



Fano_line_3_5_6.svg

0110100 · 0x34

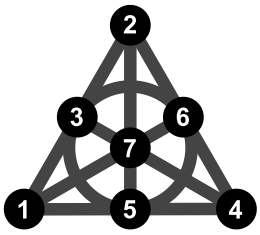
POINTS 3,5,6



Fano_line_4_7_3.svg

1001100 · 0x4c

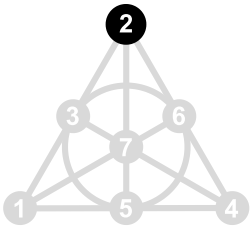
POINTS 3,4,7



Fano_full.svg

1111111 · 0x7f

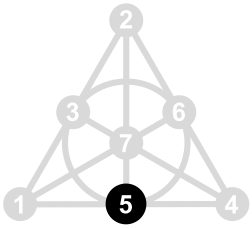
POINTS 1,2,3,4,5,6,7



Fano_point_2 (Copy).svg

0000010 · 0x02

POINTS 2 · SAME BYTES AS FANO_POINT_2.SVG



Fano_point_5 (Copy).svg

0010000 · 0x10

POINTS 5 · SAME BYTES AS FANO_POINT_5.SVG

Two views, one check

A ⊕ B shows the points where the two files differ. The count of 1s is the Hamming distance. Zero means both files hold the same constant.

VIEW A

Fano_point_2.svg

0000010

VIEW B

Fano_point_2 (Copy).svg

0000010

A ⊕ B

0000000

Hamming 0. Same constant, two views.